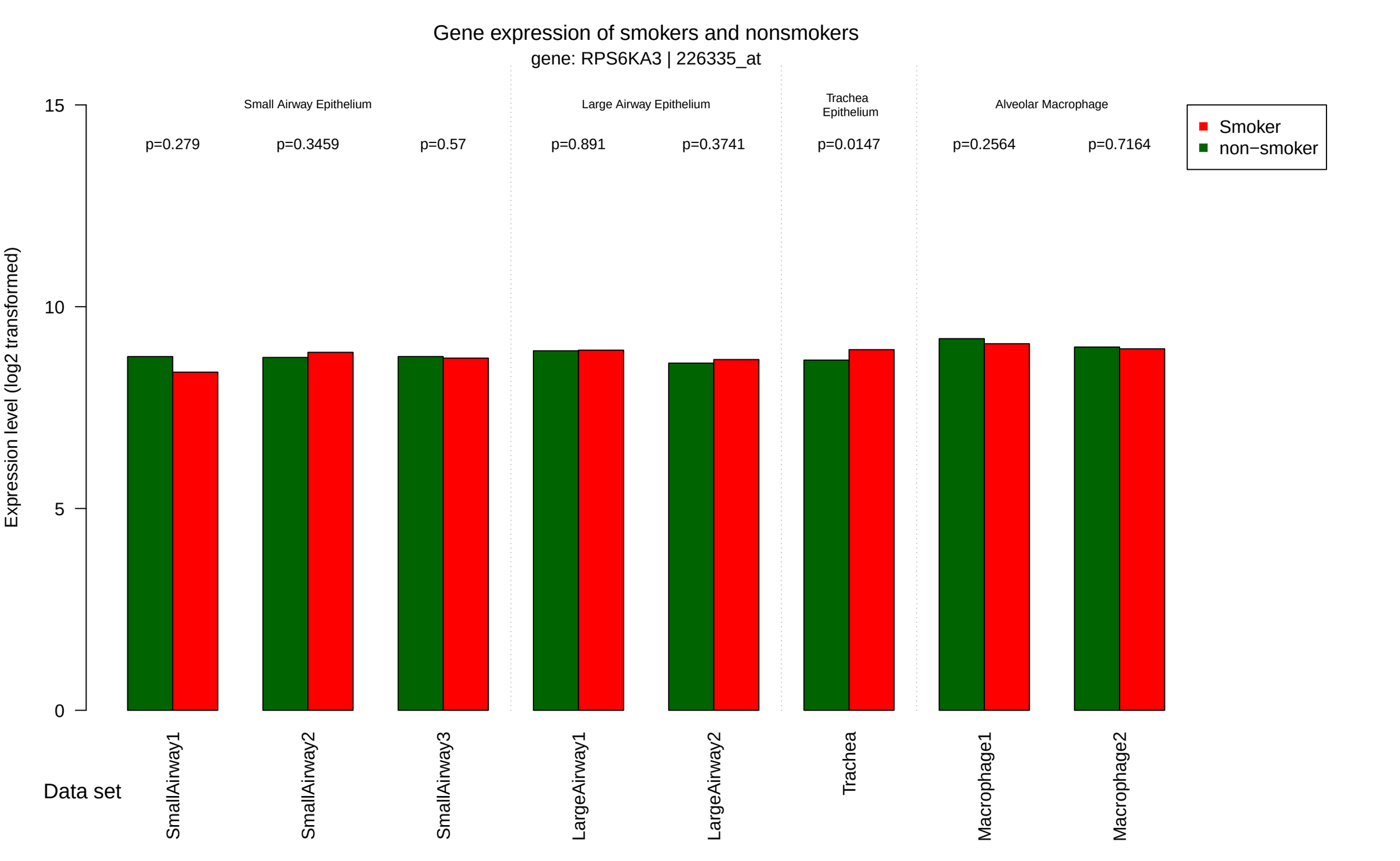
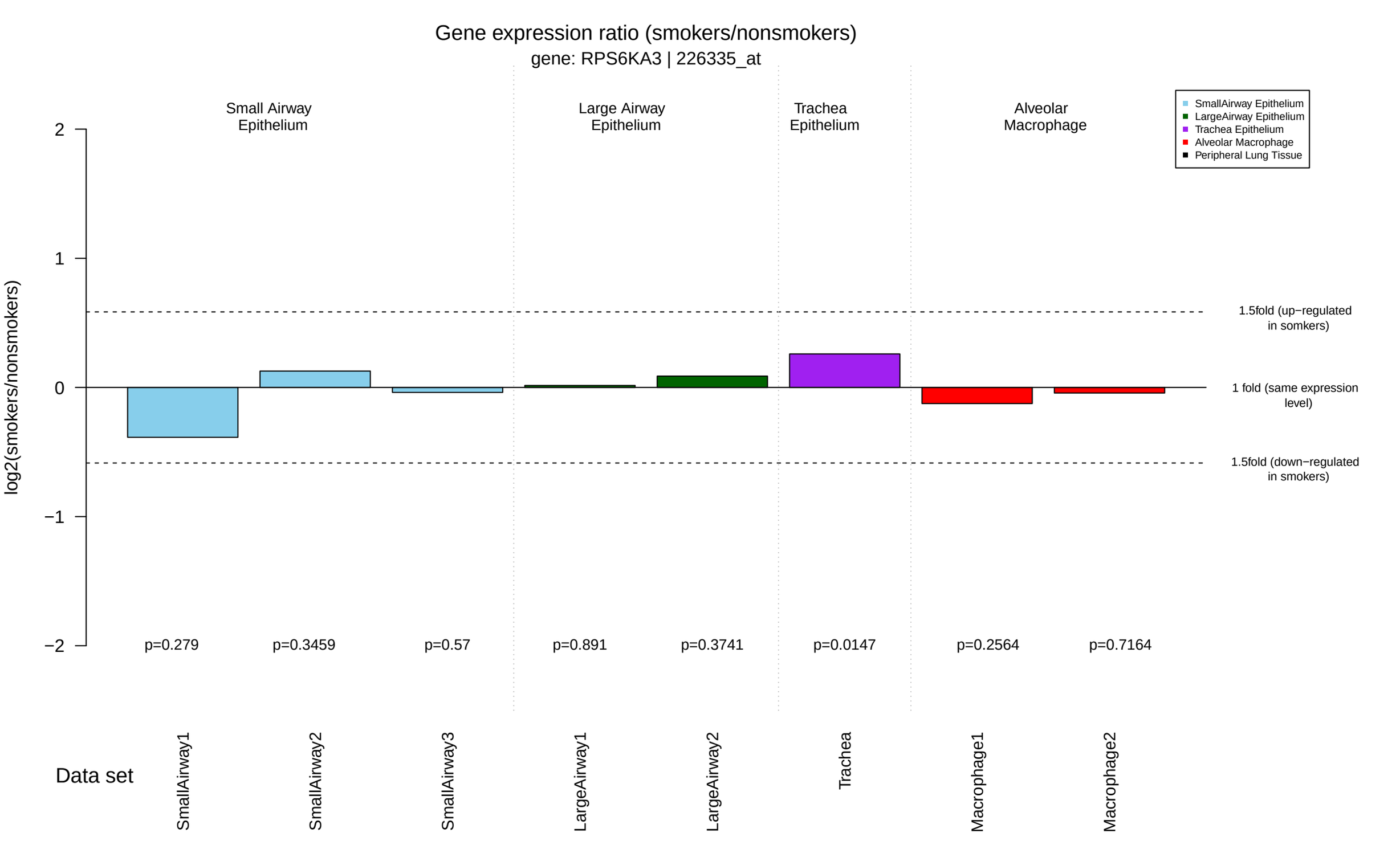
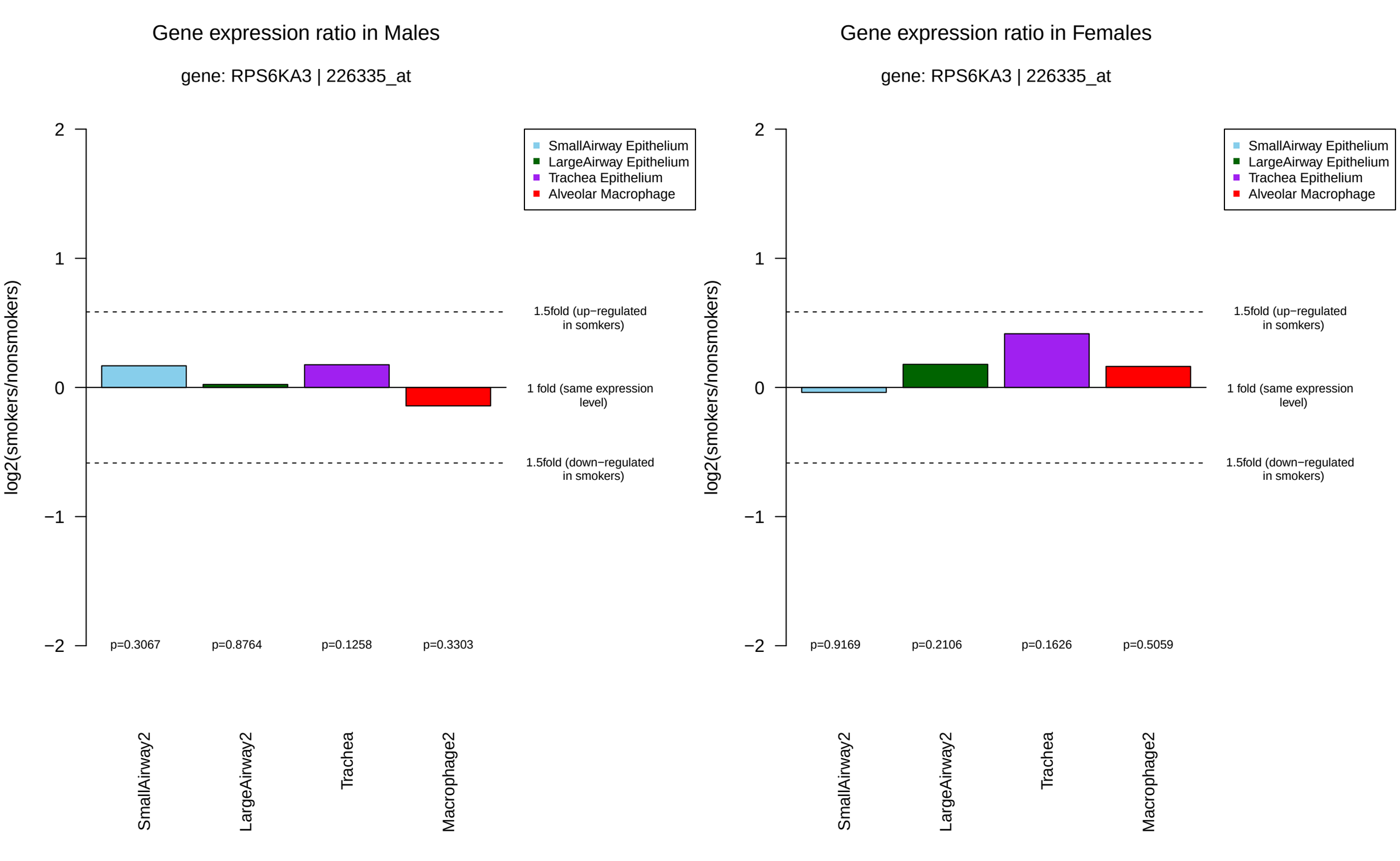
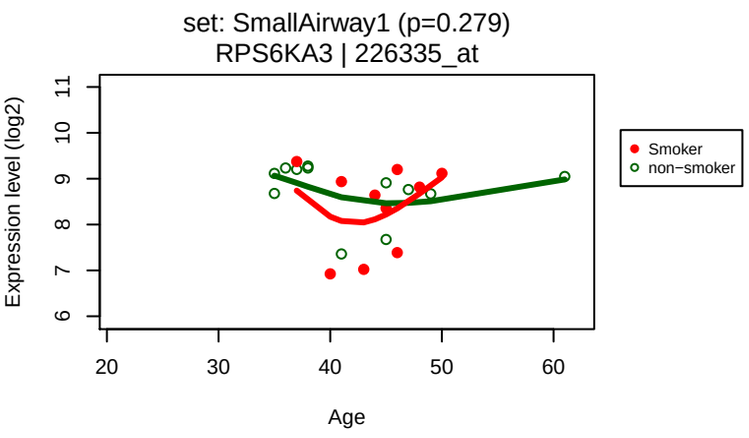
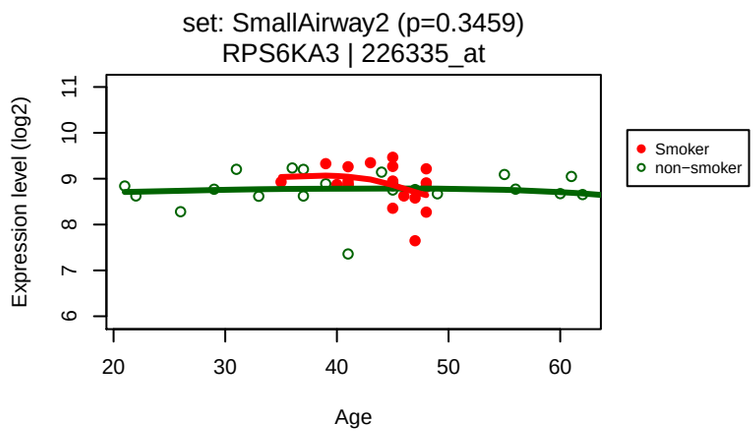
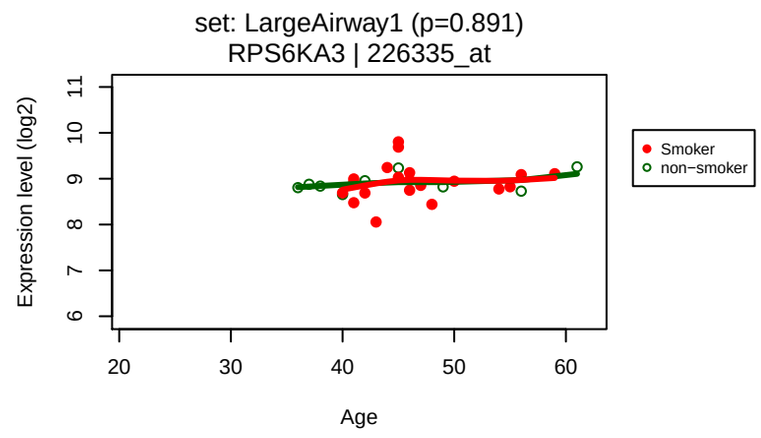

Gene expression of smokers and nonsmokers
gene: RPS6KA3 | 226335_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.279
p=0.3459
p=0.57
p=0.891
p=0.3741
p=0.0147
p=0.2564
p=0.7164
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

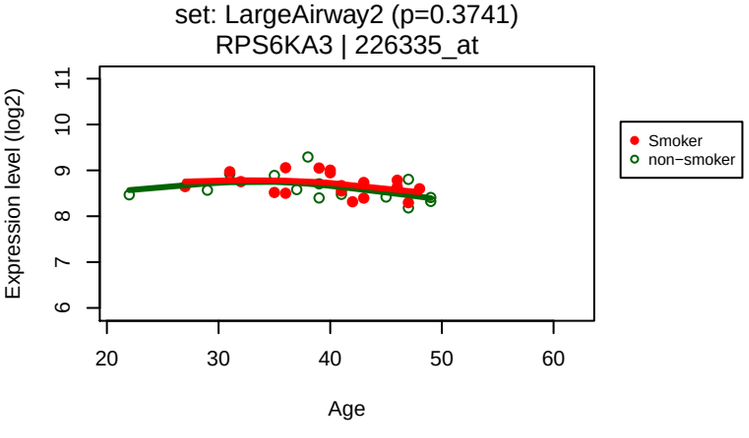

Gene expression ratio (smokers/nonsmokers)
gene: RPS6KA3 | 226335_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.279
p=0.3459
p=0.57
p=0.891
p=0.3741
p=0.0147
p=0.2564
p=0.7164
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

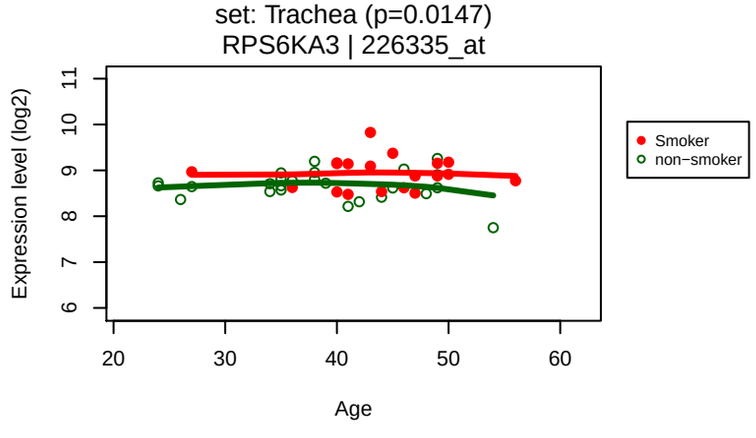

Gene expression ratio in Males
gene: RPS6KA3 | 226335_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3067
p=0.8764
p=0.1258
p=0.3303
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RPS6KA3 | 226335_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9169
p=0.2106
p=0.1626
p=0.5059
SmallAirway2
LargeAirway2
Trachea
Macrophage2

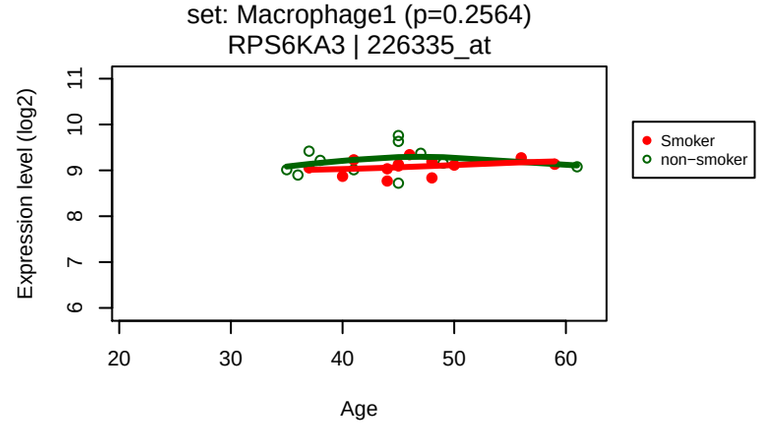

set: SmallAirway1 (p=0.279)
RPS6KA3 | 226335_at
Expression level (log2)
Age
Smoker
non−smoker

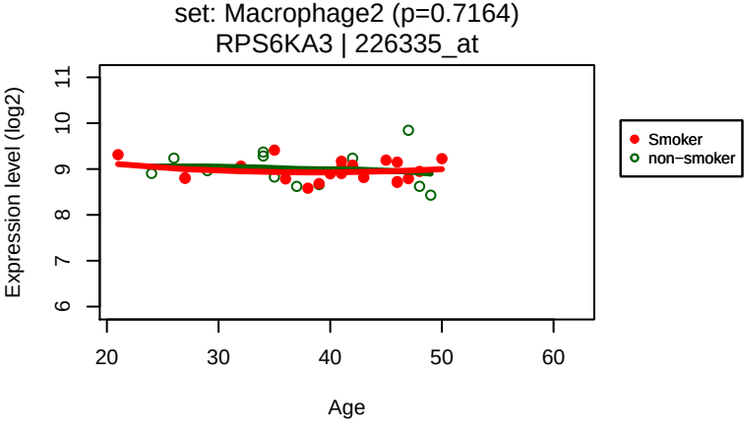

set: SmallAirway2 (p=0.3459)
RPS6KA3 | 226335_at
Expression level (log2)
Age
Smoker
non−smoker

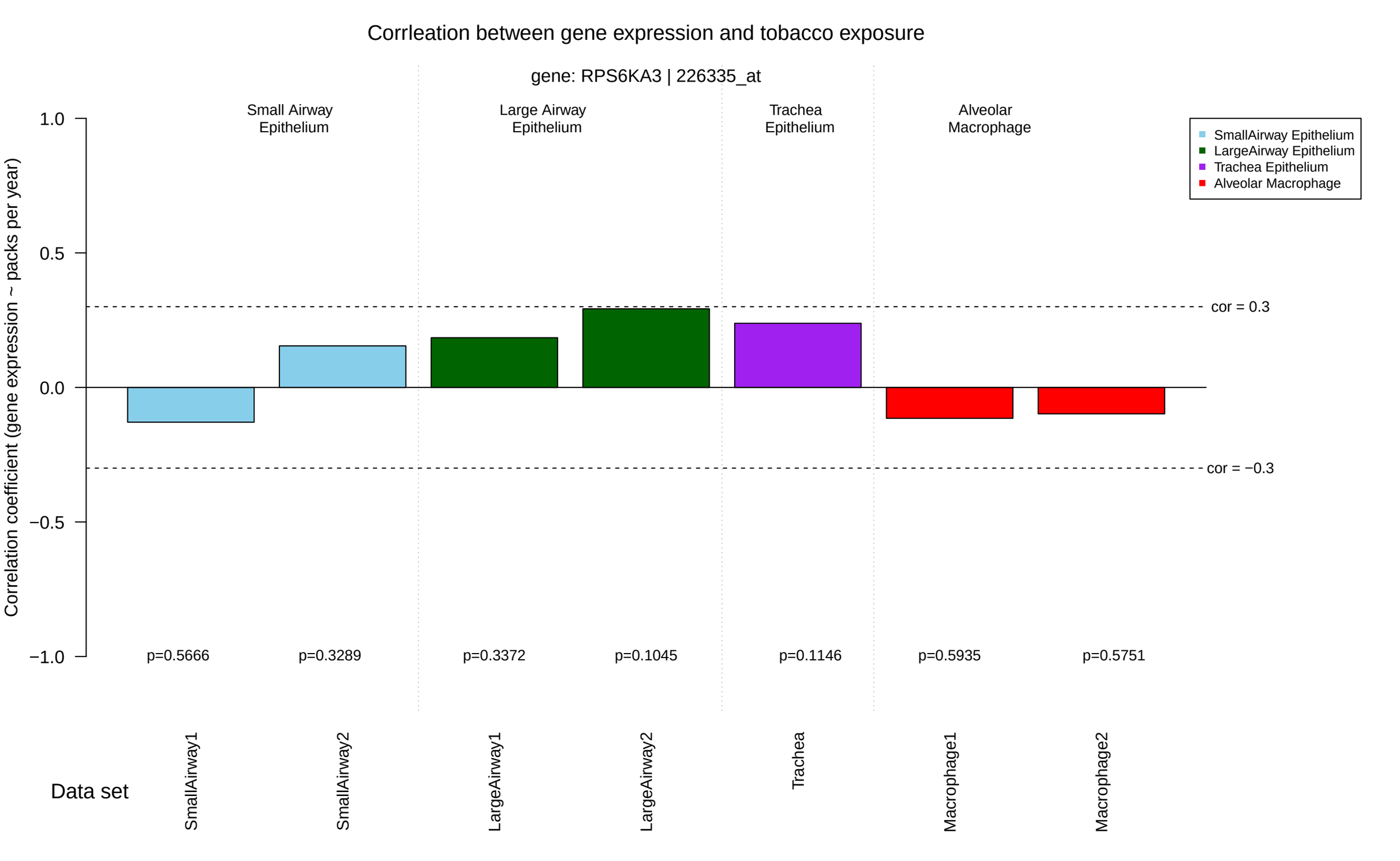

set: LargeAirway1 (p=0.891)
RPS6KA3 | 226335_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3741)
RPS6KA3 | 226335_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0147)
RPS6KA3 | 226335_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2564)
RPS6KA3 | 226335_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7164)
RPS6KA3 | 226335_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RPS6KA3 | 226335_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5666
p=0.3289
p=0.3372
p=0.1045
p=0.1146
p=0.5935
p=0.5751
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2